

Title: Plasma-Assisted Neutralization in Axial Injection for High-Current Compact Cyclotrons

Abstract High-current compact cyclotrons face strong space-charge limitations during axial injection, where the low-energy proton beam expands before reaching the inflector. We present a novel plasma-assisted neutralization module designed as a drop-in element between the final solenoid and the inflector entrance. The device consists of a short solenoid surrounding a beamline cell operated at controlled pressure. Residual hydrogen gas (H_2) or an injected noble gas (Kr) is ionized by the high-current proton beam to form a self-sustaining plasma column. Biased end-electrodes confine electrons and accelerate neutralization, achieving effective space-charge compensation. Krypton seeding provides a faster build-up of neutralization relative to H_2 , enabling rapid stabilization for pulsed or ramped operation. Simulations and analytic estimates indicate that neutralization substantially reduces transverse beam blow-up while introducing only modest multiple scattering over the short transport distance. The concept offers a compact, plasma-based method to increase capture efficiency for milliampere-class proton beams and provides a tunable platform to investigate plasma-beam interaction effects in cyclotron injection lines. Initial design parameters, integration strategy, and predicted performance are presented.